

Cash Conversion Cushion and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Cash Conversion Cushion (CCC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on CCC achieves an annualized gross (net) Sharpe ratio of 0.41 (0.38), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 37 (27) bps/month with a t-statistic of 3.56 (2.68), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price) is 37 bps/month with a t-statistic of 3.88.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While the consumption-based asset pricing paradigm provides a coherent theoretical framework for understanding risk premia, empirical identification of firm characteristics that capture consumption risk exposure remains challenging. The Cash Conversion Cushion (CCC), measuring firms' ability to convert operating activities into cash relative to their earnings, potentially captures an important dimension of consumption risk that manifests through firms' differential sensitivity to aggregate consumption shocks.

We hypothesize that the Cash Conversion Cushion proxies for firms' exposure to long-run consumption risk through multiple channels. First, firms with low cash conversion efficiency face greater sensitivity to consumption growth shocks, as their inability to generate cash from operations amplifies their exposure to aggregate demand fluctuations (Bansal and Yaron, 2004). These firms experience higher systematic risk during consumption downturns when investors' marginal utility is high, requiring higher expected returns as compensation (Campbell and Cochrane, 1999). The state-dependent nature of this risk exposure becomes particularly pronounced during economic contractions when credit constraints bind and internal cash generation becomes critical for survival (Jagannathan and Wang, 1996).

Second, the cash conversion mechanism relates directly to the intertemporal marginal rate of substitution through its impact on firms' investment and dividend policies. Firms with weak cash conversion face greater adjustment costs in responding to consumption shocks, creating time-varying risk exposure that correlates with the pricing kernel (Zhang, 2005). This friction in capital reallocation amplifies the covariance between firm cash flows and marginal utility growth, particularly in bad states when consumption disasters become more probable (Barro, 2006). The result-

ing risk premium reflects compensation for bearing undiversifiable consumption risk that varies systematically with macroeconomic conditions.

Third, cash conversion efficiency determines firms' resilience to habit-based consumption risk. When consumption approaches the habit level, investors become increasingly risk-averse, and firms with poor cash conversion face disproportionate valuation impacts (Campbell and Cochrane, 1999). The mechanism operates through the interaction between operating leverage and consumption smoothing: firms unable to efficiently convert earnings to cash cannot provide the consumption hedging that risk-averse investors demand during periods of heightened marginal utility (Santos and Veronesi, 2000). This creates persistent cross-sectional variation in required returns that aligns with consumption-based asset pricing predictions.

The Cash Conversion Cushion strategy generates economically and statistically significant returns across multiple specifications. The value-weighted long-short portfolio that buys high CCC stocks and sells low CCC stocks earned an average monthly return of 34 basis points with a t-statistic of 3.10 over the 1967-2024 sample period. The strategy's annualized Sharpe ratio of 0.41 places it in the 82nd percentile among 212 documented anomalies, demonstrating strong risk-adjusted performance. Most notably, the strategy maintains significant alpha of 37 basis points per month (t-statistic = 3.56) after controlling for the Fama-French five factors plus momentum, indicating that CCC captures a dimension of systematic risk not explained by existing factors.

The economic significance of the CCC premium persists after accounting for transaction costs and remains robust across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average return of 32 basis points per month with a t-statistic of 2.86, and maintains a significant generalized alpha of 27 basis points (t-statistic = 2.68) relative to the six-factor model. The strategy's net Sharpe ratio of 0.38 ranks in the 94th percentile of the anomaly distribution, sug-

gesting that CCC provides an implementable investment opportunity. Among the largest quintile of stocks by market capitalization, the strategy continues to generate significant returns of 28 basis points per month (t -statistic = 2.12), demonstrating that the effect is not confined to small, illiquid securities.

The CCC strategy exhibits distinct factor loadings consistent with consumption risk exposure. The strategy loads positively on the market factor ($\beta = 0.18$, t -statistic = 7.43) and negatively on the profitability factor ($\beta = -0.28$, t -statistic = -5.84), reflecting the consumption risk channel through which less profitable firms with poor cash conversion face higher systematic risk. Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the strategy maintains an alpha of 37 basis points per month with a t -statistic of 3.88, confirming that CCC captures unique information about the cross-section of expected returns not subsumed by existing predictors.

This paper contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to systematic consumption risk. While [Bansal and Yaron \(2004\)](#) establish the theoretical importance of long-run consumption risk and [Parker and Julliard \(2005\)](#) provide empirical evidence linking consumption to asset prices, we demonstrate that cash conversion efficiency serves as an observable proxy for firms' consumption risk exposure. Our findings extend [Jagannathan and Wang \(1996\)](#) by showing that operational efficiency metrics contain information about systematic risk beyond traditional accounting measures. The CCC signal provides a direct link between firm fundamentals and the stochastic discount factor, offering empirical support for consumption-based theories of the cross-section.

Methodologically, we advance the literature on anomaly discovery by subjecting the CCC signal to comprehensive robustness tests following [Harvey et al. \(2016\)](#). Unlike many documented anomalies that fail to survive transaction costs or are con-

centrated among micro-cap stocks, the CCC strategy generates significant net returns across all size quintiles and under various portfolio construction methods. Our spanning tests demonstrate that CCC expands the mean-variance efficient frontier even after controlling for 156 existing anomalies, addressing concerns raised by [Cochrane \(2011\)](#) about factor proliferation. The strategy’s performance relative to the entire factor zoo, achieving top-decile net Sharpe ratios, suggests that CCC captures a fundamental risk dimension rather than a spurious statistical pattern.

Our results have important implications for understanding the economic mechanisms underlying cross-sectional return variation. The persistence of the CCC premium across different market conditions and its alignment with consumption-based pricing kernels support rational risk explanations over behavioral interpretations. By linking operational efficiency to required returns through the consumption risk channel, we provide a unified framework for understanding why certain firm characteristics predict returns. This insight informs both asset pricing theory, by validating consumption-based models with micro-level data, and investment practice, by identifying an implementable strategy that captures systematic risk exposure. The robustness of our findings to alternative specifications and the economic magnitude of the premium suggest that cash conversion efficiency represents a fundamental determinant of equilibrium expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of cash holdings and operating perfor-

mance metrics across firms. signal construction relies on two key financial statement variables. Cash (CH) represents the firm’s cash and short-term investments as reported on the balance sheet, capturing the most liquid assets readily available to the firm. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, representing the firm’s operating performance before the effects of capital structure, tax regime, and non-cash accounting charges. This metric provides a cleaner measure of cash-generating ability from core operations compared to net income. construct the Cash Conversion Cushion signal by dividing cash and short-term investments (CH) by EBITDA (EBITDA). This ratio captures the relationship between a firm’s liquid asset buffer and its operating cash generation capacity, effectively measuring how many years of current operating performance the firm’s cash holdings represent. We calculate this signal using fiscal year-end values to ensure temporal consistency across firms with different fiscal year endings. In our signal construction, we require non-missing values for both CH and EBITDA, and we exclude observations where EBITDA is zero or negative to avoid undefined or economically meaningless ratios. The resulting measure provides insight into a firm’s financial flexibility relative to its operating scale, with higher values indicating a larger cash cushion relative to operating performance.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the CCC signal. Panel A plots the time-series of the mean, median, and interquartile range for CCC. On average, the cross-sectional mean (median) CCC is 0.42 (0.29) over the 1967 to 2024 sample, where the starting date is determined by the availability of the input CCC data. The signal’s interquartile range spans -0.96 to 1.43. Panel B of Figure 1 plots the time-series of the coverage of the CCC signal for the CRSP universe. On average, the CCC signal

is available for 6.58% of CRSP names, which on average make up 7.06% of total market capitalization.

4 Does CCC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on CCC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high CCC portfolio and sells the low CCC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short CCC strategy earns an average return of 0.34% per month with a t-statistic of 3.10. The annualized Sharpe ratio of the strategy is 0.41. The alphas range from 0.18% to 0.37% per month and have t-statistics exceeding 1.79 everywhere. The lowest alpha is with respect to the FF3 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 7.43 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 462 stocks and an average market capitalization of at least \$1,310 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and value-weighted portfolios, and equals 34 bps/month with a t-statistics of 3.10. Out of the twenty-five alphas reported in Panel A, the t-statistics for fourteen exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.27. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the CCC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-one cases, and significantly expands the achievable frontier in nine cases.

Table 3 provides direct tests for the role size plays in the CCC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and CCC, as well as average returns and alphas for long/short trading CCC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the CCC strategy achieves an average return of 28 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the CCC strategy relative to the five most common factor models range from 12 to 41 bps/month with t-statistics between 0.99 and 3.30.

5 How does CCC perform relative to the zoo?

Figure 2 puts the performance of CCC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the CCC strategy falls in the distribution. The CCC strategy’s gross (net) Sharpe ratio of 0.41 (0.38) is greater than 82% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the CCC strategy (red line).² Ignoring trading costs, a \$1 invested in the CCC strategy would have yielded \$6.03 which ranks the CCC strategy in the top 3% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the CCC strategy would have yielded \$4.85 which ranks the CCC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the CCC relative to those. Panel A shows that the CCC strategy gross alphas fall between the 45 and 85 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196706 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The CCC strategy has a positive net generalized alpha for five out of the five factor models. In these cases CCC ranks between the 62 and 93 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does CCC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of CCC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price CCC or at least to weaken the power CCC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on CCC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the CCC signal in these Fama-MacBeth regressions exceed -0.86, with the minimum t-statistic occurring when controlling for Idiosyncratic risk (AHT). Controlling for all six closely related anomalies, the t-statistic on CCC is

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

3.11.

Similarly, Table 5 reports results from spanning tests that regress returns to the CCC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the CCC strategy earns alphas that range from 33-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.22, which is achieved when controlling for Idiosyncratic risk (AHT). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the CCC trading strategy achieves an alpha of 37bps/month with a t-statistic of 3.88.

7 Does CCC add relative to the whole zoo?

Finally, we can ask how much adding CCC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 156 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 156 anomalies augmented with the CCC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 156-anomaly combination strategy grows to \$2068.13, while \$1 investment in the combination strategy that includes CCC grows to \$2287.89.

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which CCC is available.

8 Conclusion

This study provides compelling evidence that the Cash Conversion Cushion (CCC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that CCC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on CCC delivers an annualized Sharpe ratio of 0.41 gross (0.38 net of transaction costs), indicating strong performance that remains economically meaningful after accounting for implementation costs. Most notably, the strategy generates a monthly alpha of 37 basis points (t -statistic = 3.56) relative to the Fama-French five-factor model augmented with momentum, declining modestly to 27 basis points net of costs while maintaining statistical significance (t -statistic = 2.68). The robustness of these results is further confirmed by the persistence of a 37 basis point monthly alpha (t -statistic = 3.88) even after controlling for six closely related strategies from the factor zoo, including analyst earnings forecasts, idiosyncratic risk measures, profitability metrics, and financing activities.

These findings have important practical implications for both academic research and investment practice. The CCC signal appears to capture a distinct dimension of firm performance related to working capital efficiency and cash flow management that is not fully subsumed by existing factor models or related anomalies. For practitioners, the strategy’s resilience to transaction costs and its orthogonality to established factors suggest potential portfolio diversification benefits and the possibility of enhanced risk-adjusted returns through the incorporation of CCC-based screens in investment processes.

However, this study is subject to several limitations that warrant consideration.

First, while we employ conservative transaction cost estimates, actual implementation costs may vary depending on market conditions, trade execution capabilities, and portfolio size. Second, the analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the CCC anomaly require further investigation to determine whether the premium reflects compensation for systematic risk, market inefficiencies, or behavioral biases.

Future research should explore several promising avenues. International evidence would help establish whether the CCC effect represents a global phenomenon or is specific to U.S. market structures. Time-series analysis of the CCC premium could reveal whether its predictive power varies with business cycles, market conditions, or changes in accounting standards. Additionally, investigating the interaction between CCC and corporate events such as earnings announcements, management changes, or strategic initiatives could provide deeper insights into the economic drivers of the anomaly. Finally, examining the relationship between CCC and firm-level outcomes such as default risk, profitability persistence, and investment efficiency would enhance our understanding of why this signal predicts future returns and whether it represents a genuine risk factor deserving inclusion in asset pricing models.

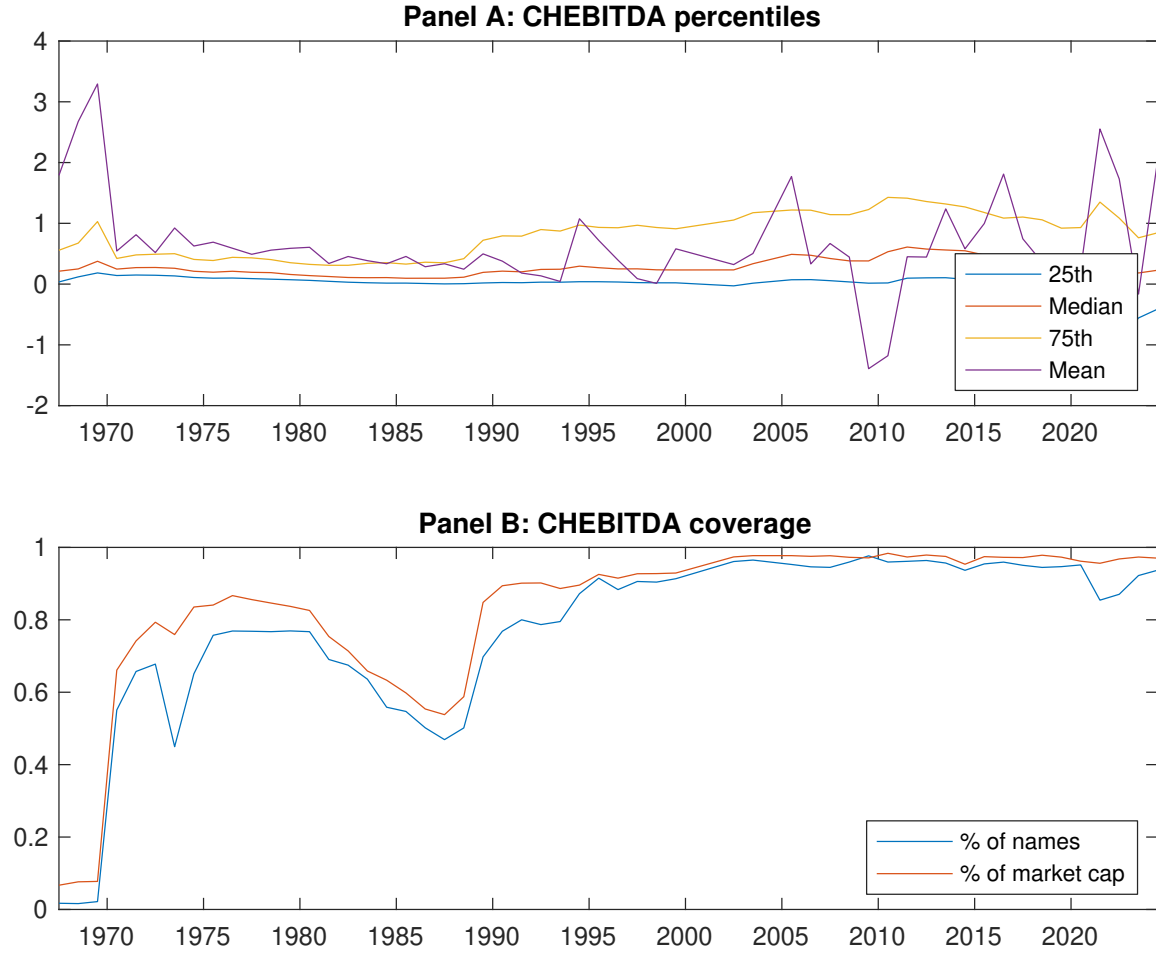


Figure 1: Times series of CCC percentiles and coverage.
This figure plots descriptive statistics for CCC. Panel A shows cross-sectional percentiles of CCC over the sample. Panel B plots the monthly coverage of CCC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on CCC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Excess returns and alphas on CCC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.59]	0.43 [2.37]	0.55 [2.91]	0.53 [2.57]	0.79 [3.68]	0.34 [3.10]
α_{CAPM}	-0.09 [-1.46]	-0.12 [-1.58]	-0.04 [-0.61]	-0.11 [-1.53]	0.11 [1.53]	0.20 [1.97]
α_{FF3}	-0.08 [-1.29]	-0.14 [-1.91]	-0.07 [-1.03]	-0.11 [-1.54]	0.10 [1.45]	0.18 [1.79]
α_{FF4}	-0.11 [-1.75]	-0.13 [-1.73]	-0.10 [-1.37]	-0.09 [-1.16]	0.15 [2.12]	0.26 [2.54]
α_{FF5}	-0.05 [-0.84]	-0.28 [-3.89]	-0.16 [-2.28]	-0.15 [-1.94]	0.26 [3.79]	0.31 [3.00]
α_{FF6}	-0.08 [-1.26]	-0.26 [-3.60]	-0.17 [-2.46]	-0.12 [-1.60]	0.29 [4.21]	0.37 [3.56]
Panel B: Fama and French (2018) 6-factor model loadings for CCC-sorted portfolios						
β_{MKT}	0.92 [58.96]	1.00 [57.78]	1.05 [63.13]	1.08 [59.41]	1.10 [68.09]	0.18 [7.43]
β_{SMB}	0.02 [0.77]	-0.05 [-1.88]	0.06 [2.61]	0.11 [4.25]	0.08 [3.22]	0.06 [1.62]
β_{HML}	0.02 [0.60]	0.01 [0.17]	0.04 [1.19]	-0.04 [-1.07]	0.07 [2.20]	0.05 [1.06]
β_{RMW}	-0.05 [-1.56]	0.34 [10.14]	0.19 [5.97]	0.11 [3.01]	-0.33 [-10.40]	-0.28 [-5.84]
β_{CMA}	-0.08 [-1.69]	0.10 [2.10]	0.08 [1.70]	0.02 [0.30]	-0.15 [-3.22]	-0.07 [-1.04]
β_{UMD}	0.04 [2.75]	-0.03 [-1.56]	0.02 [1.36]	-0.04 [-2.04]	-0.05 [-2.86]	-0.09 [-3.62]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1011	462	524	636	1003	
me (\$10 ⁶)	1310	2472	2653	2310	1846	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the CCC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196706 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [3.10]	0.20 [1.97]	0.18 [1.79]	0.26 [2.54]	0.31 [3.00]	0.37 [3.56]
Quintile	NYSE	EW	0.36 [2.77]	0.44 [3.37]	0.34 [2.81]	0.26 [2.13]	0.12 [1.05]	0.07 [0.58]
Quintile	Name	VW	0.37 [2.20]	0.43 [2.60]	0.26 [1.74]	0.29 [1.88]	-0.06 [-0.40]	-0.01 [-0.10]
Quintile	Cap	VW	0.38 [3.42]	0.24 [2.33]	0.27 [2.58]	0.32 [3.08]	0.43 [4.13]	0.46 [4.42]
Decile	NYSE	VW	0.42 [3.10]	0.36 [2.65]	0.26 [1.94]	0.33 [2.46]	0.20 [1.50]	0.27 [1.97]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.32 [2.86]	0.16 [1.54]	0.14 [1.39]	0.19 [1.85]	0.24 [2.29]	0.27 [2.68]
Quintile	NYSE	EW	0.17 [1.27]	0.24 [1.79]	0.14 [1.13]	0.10 [0.80]		
Quintile	Name	VW	0.32 [1.92]	0.37 [2.21]	0.20 [1.37]	0.22 [1.46]		
Quintile	Cap	VW	0.35 [3.15]	0.20 [1.91]	0.22 [2.12]	0.25 [2.44]	0.34 [3.32]	0.36 [3.54]
Decile	NYSE	VW	0.39 [2.84]	0.32 [2.34]	0.22 [1.68]	0.26 [1.99]	0.17 [1.22]	0.21 [1.59]

Table 3: Conditional sort on size and CCC

This table presents results for conditional double sorts on size and CCC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on CCC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high CCC and short stocks with low CCC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196706 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	CCC Quintiles					CCC Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.30 [0.89]	0.51 [1.81]	0.67 [2.59]	0.81 [3.27]	0.93 [3.42]	0.62 [3.09]	0.77 [3.87]	0.60 [3.19]	0.53 [2.82]	0.24 [1.35]	0.21 [1.17]
	(2)	0.47 [1.62]	0.73 [3.00]	0.70 [2.91]	0.74 [3.09]	0.68 [2.62]	0.33 [2.02]	0.42 [2.58]	0.26 [1.70]	0.25 [1.60]	0.07 [0.44]	0.06 [0.43]
	(3)	0.51 [2.07]	0.55 [2.39]	0.82 [3.57]	0.65 [2.63]	0.63 [2.75]	0.09 [0.70]	0.11 [0.86]	0.02 [0.15]	-0.05 [-0.41]	-0.11 [-0.86]	-0.16 [-1.25]
	(4)	0.46 [2.21]	0.51 [2.33]	0.50 [2.25]	0.78 [3.54]	0.85 [3.61]	0.39 [2.98]	0.30 [2.30]	0.31 [2.44]	0.32 [2.46]	0.42 [3.21]	0.42 [3.20]
	(5)	0.47 [2.80]	0.44 [2.53]	0.40 [2.11]	0.48 [2.40]	0.75 [3.48]	0.28 [2.12]	0.12 [0.99]	0.14 [1.09]	0.22 [1.73]	0.36 [2.88]	0.41 [3.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	CCC Quintiles					CCC Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	412	409	420	422	422	31	32	40	42	39	
	(2)	117	112	117	117	113	62	62	66	67	63	
	(3)	79	78	83	80	78	106	107	115	109	105	
	(4)	63	63	63	64	63	217	225	224	225	226	
(5)	56	56	56	56	56	1248	2023	1990	1710	1475		

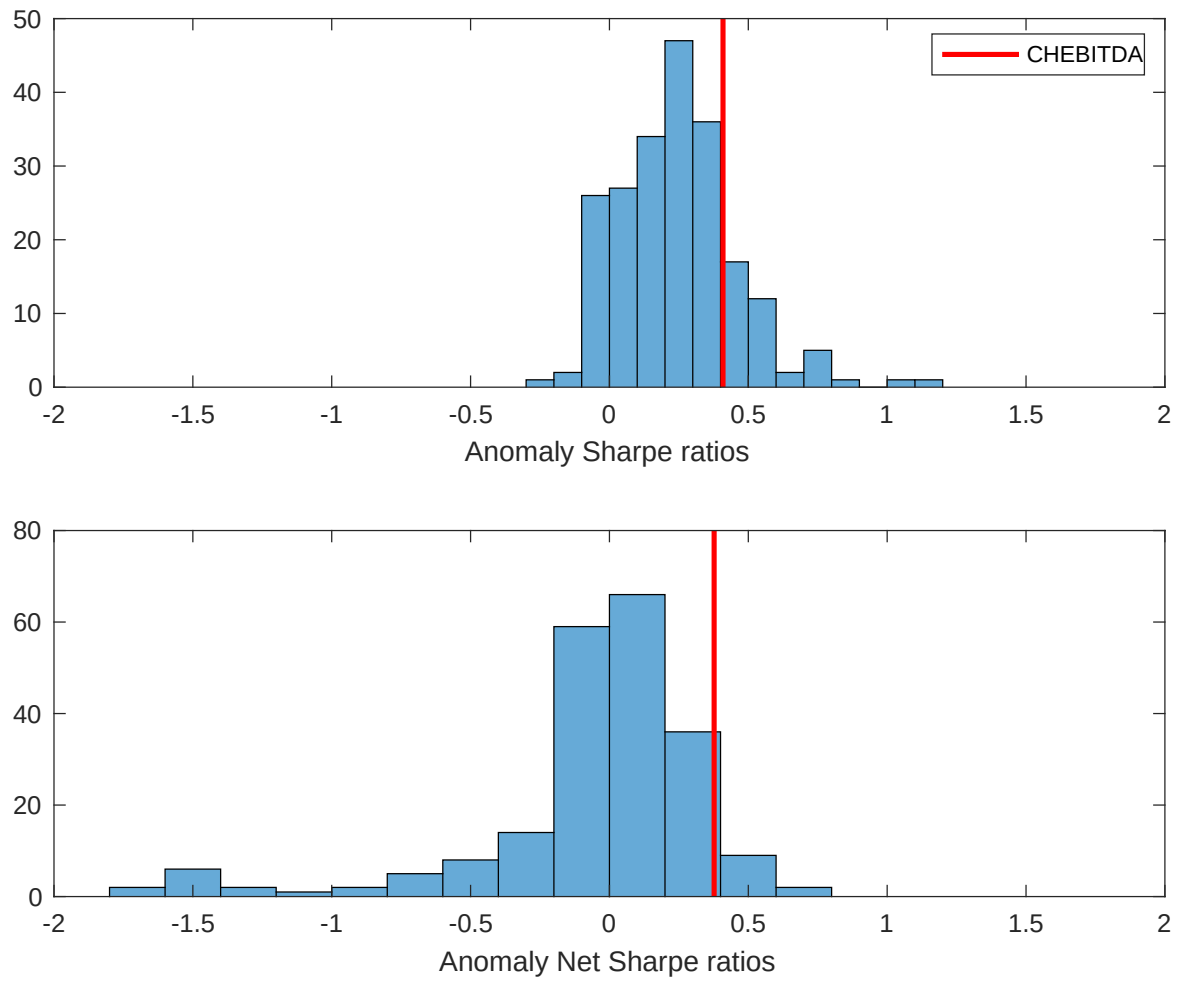


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the CCC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

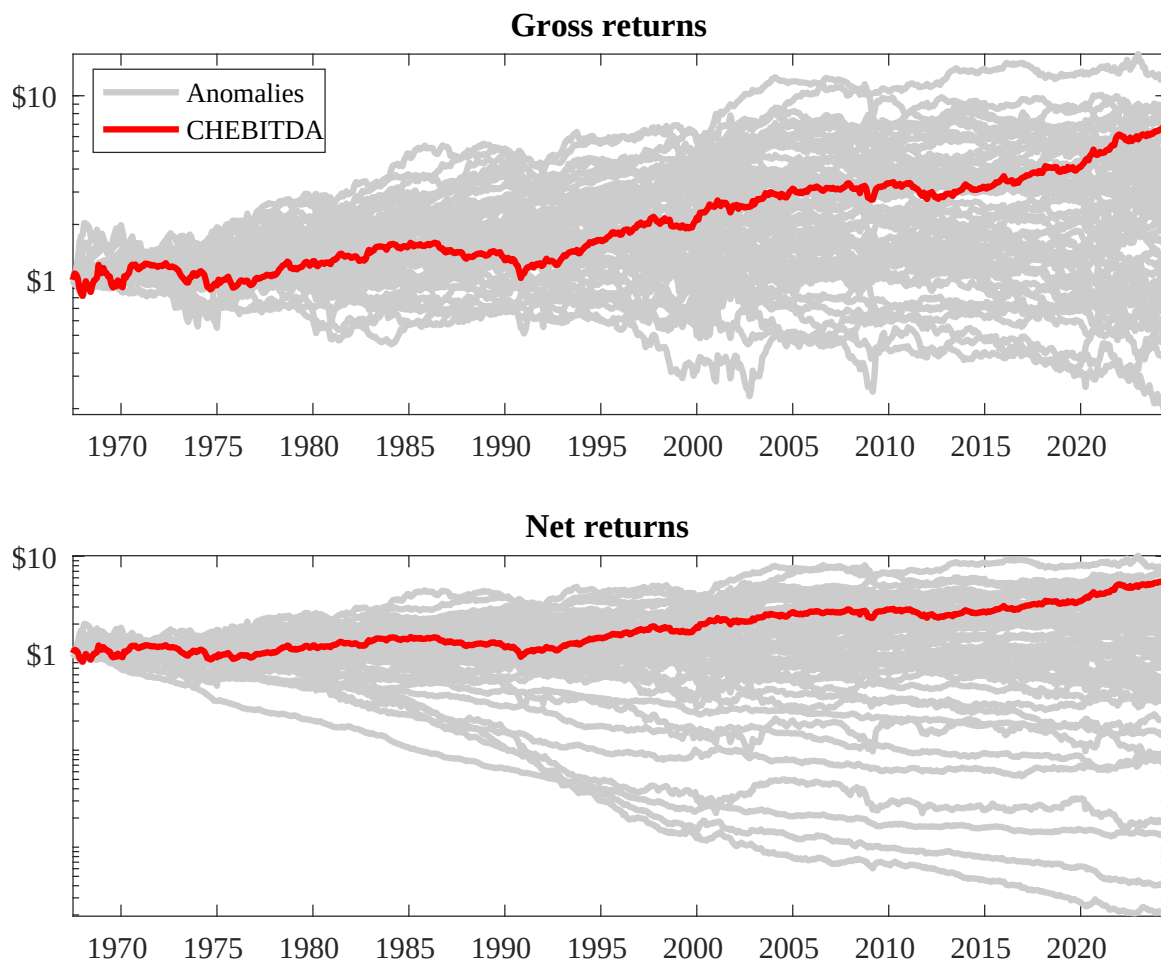
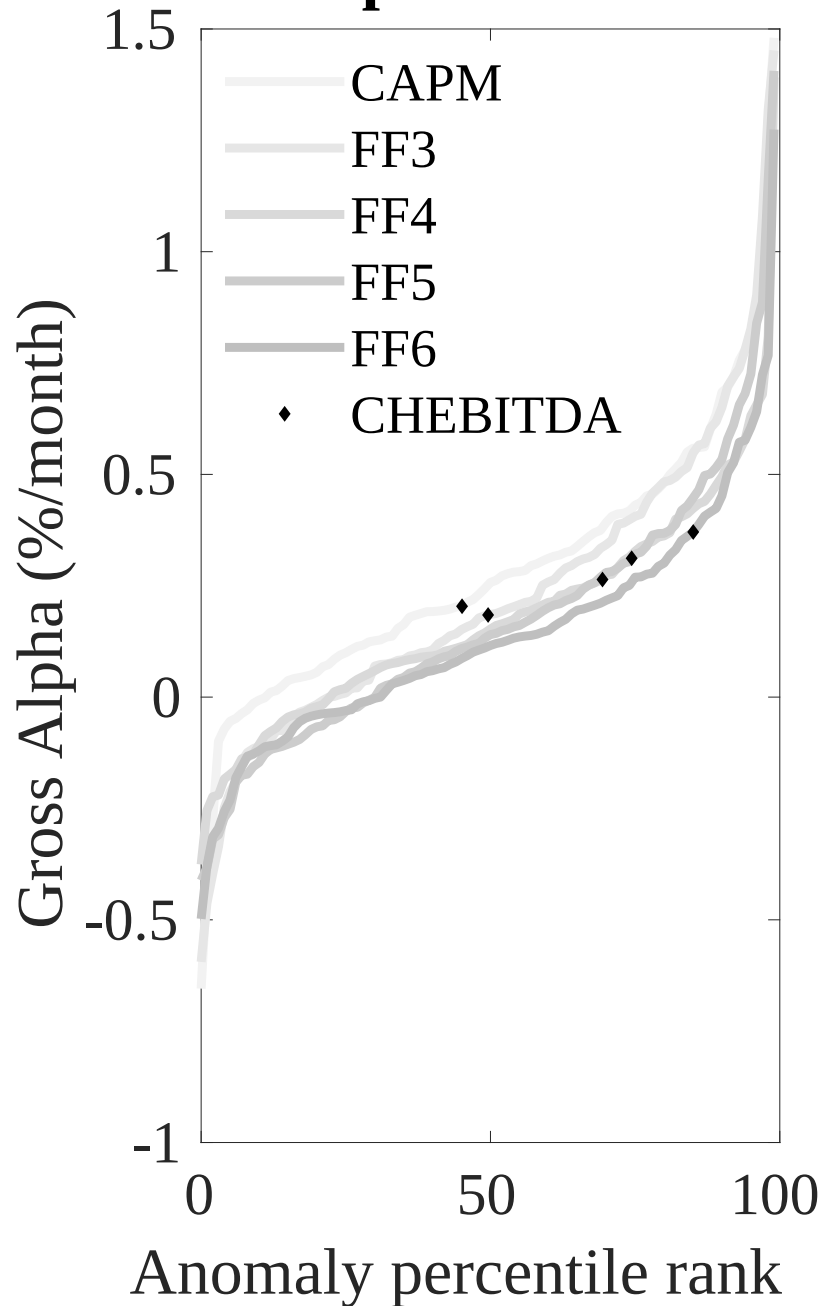


Figure 3: Dollar invested.

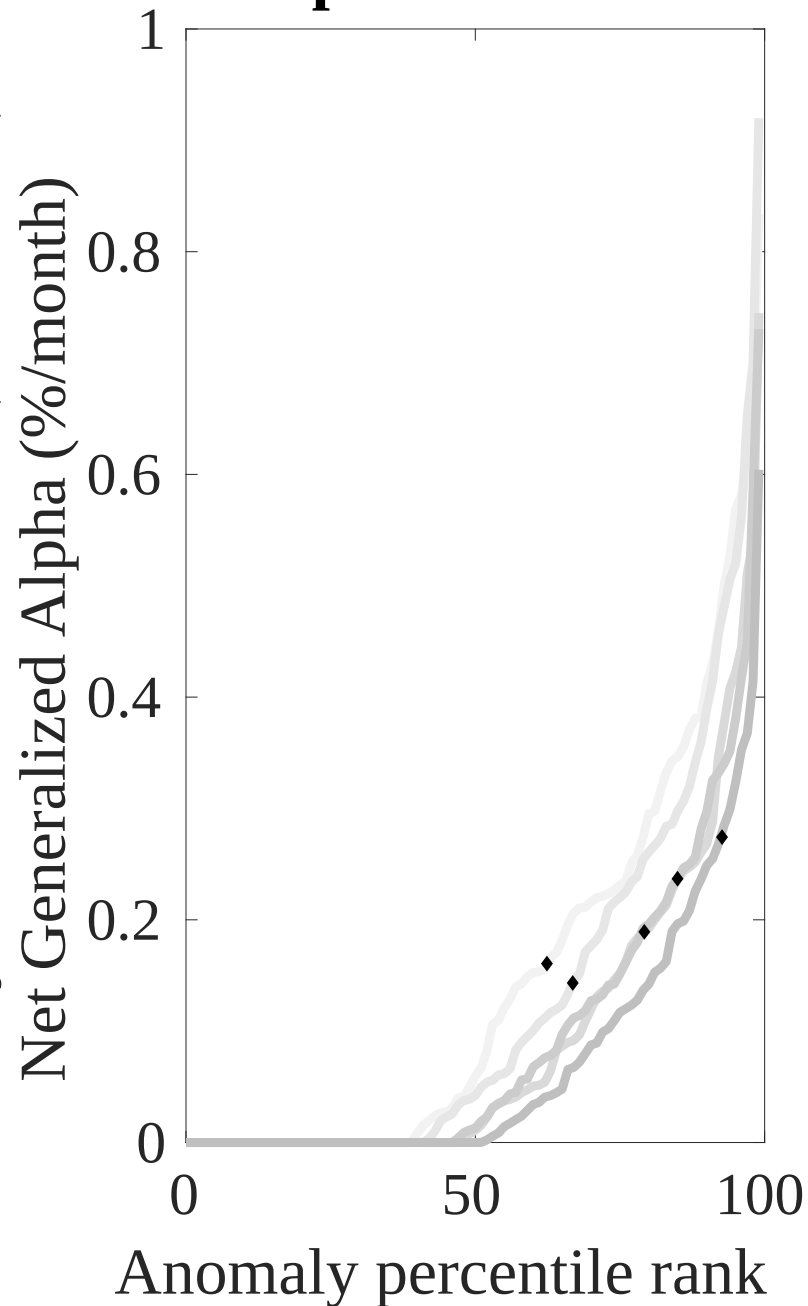
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the CCC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



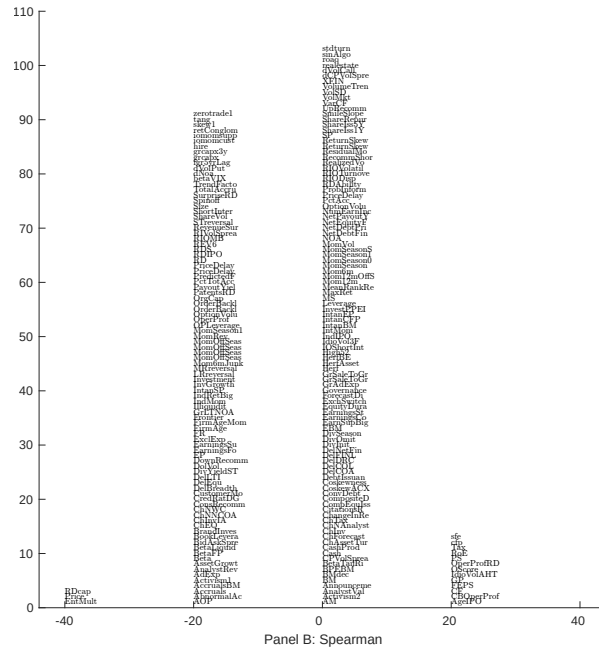
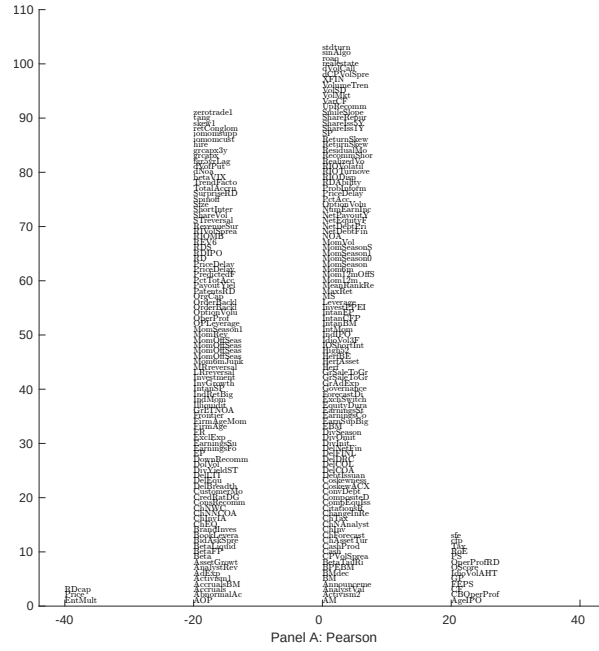


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with CCC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

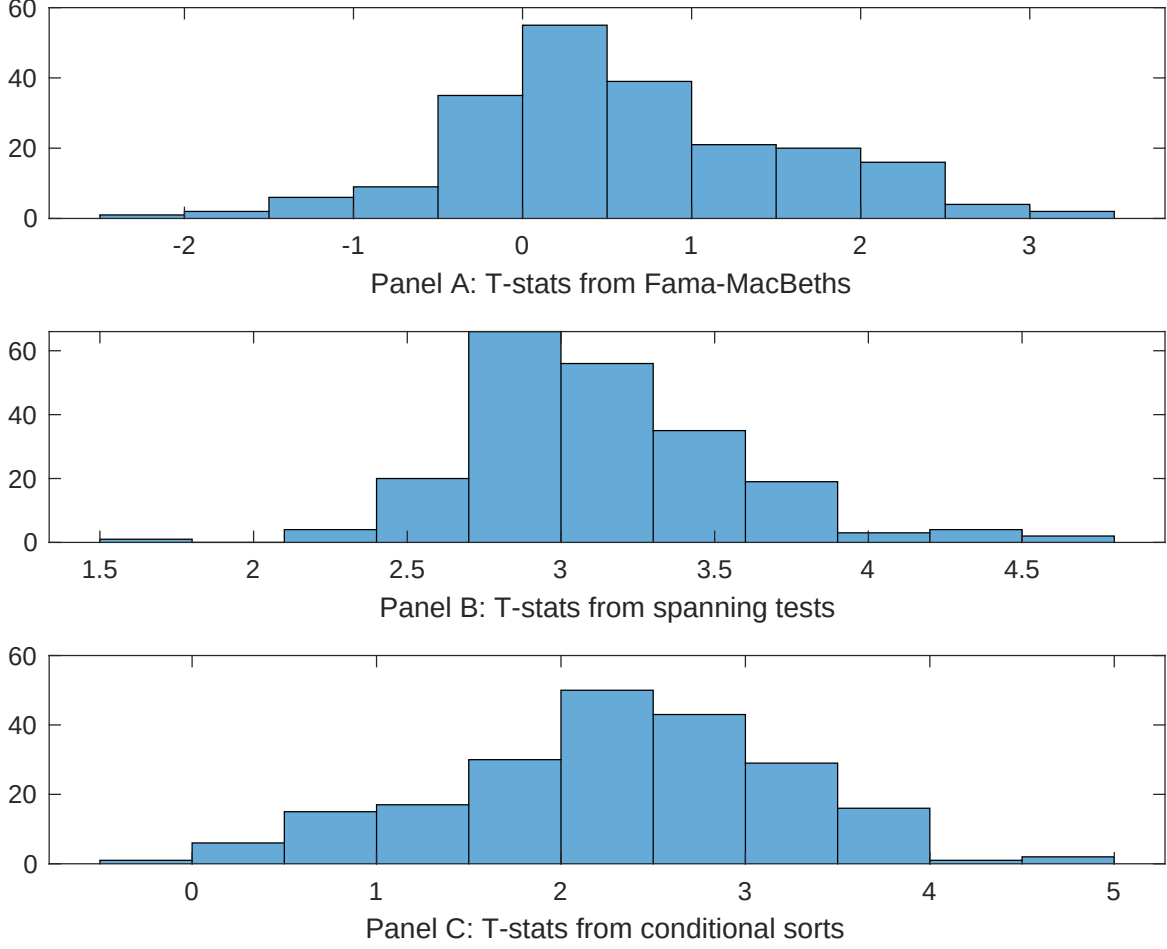


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of CCC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{CCC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{CCC} CCC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{CCC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on CCC. Stocks are finally grouped into five CCC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted CCC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on CCC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{CCC}CCC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.97 [3.37]	0.14 [7.49]	0.12 [4.94]	0.13 [5.31]	0.15 [8.01]	0.12 [4.28]	0.16 [7.57]
CCC	0.58 [0.60]	-0.65 [-0.86]	-0.32 [-0.03]	0.22 [1.97]	-0.63 [-0.77]	0.14 [0.02]	0.55 [3.11]
Anomaly 1	0.84 [1.51]						0.12 [3.46]
Anomaly 2		0.95 [1.71]					-0.82 [-1.33]
Anomaly 3			0.47 [0.04]				0.10 [1.13]
Anomaly 4				0.19 [7.13]			0.12 [5.68]
Anomaly 5					0.16 [3.58]		0.27 [9.94]
Anomaly 6						0.61 [1.29]	0.17 [4.74]
# months	576	643	648	630	643	648	576
$\bar{R}^2(\%)$	2	2	0	1	2	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the CCC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{CCC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Analyst earnings per share, Idiosyncratic risk (AHT), net income / book equity, Net external financing, Idiosyncratic risk (3 factor), Price. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196706 to 202406.

Intercept	0.37 [3.88]	0.34 [3.22]	0.36 [3.41]	0.33 [3.66]	0.34 [3.25]	0.36 [3.41]	0.37 [3.88]
Anomaly 1	4.75 [1.26]						7.58 [1.73]
Anomaly 2		1.01 [0.28]					-4.64 [-0.92]
Anomaly 3			-9.23 [-1.57]				-7.34 [-1.21]
Anomaly 4				2.46 [0.54]			9.37 [1.75]
Anomaly 5					-1.62 [-0.44]		-4.19 [-0.86]
Anomaly 6						-4.46 [-1.02]	-3.01 [-0.69]
mkt	18.46 [7.77]	18.85 [6.67]	17.01 [6.54]	17.51 [7.88]	17.90 [6.47]	19.05 [7.49]	16.52 [6.32]
smb	12.88 [3.04]	7.78 [1.53]	3.48 [0.86]	12.44 [3.56]	5.41 [1.13]	11.17 [1.90]	10.68 [1.95]
hml	6.73 [1.54]	5.52 [1.13]	4.79 [0.99]	8.46 [2.07]	6.28 [1.28]	6.42 [1.33]	9.47 [2.00]
rmw	-22.49 [-3.90]	-28.97 [-4.93]	-19.53 [-2.70]	-20.57 [-4.15]	-26.84 [-4.86]	-30.47 [-5.68]	-18.37 [-2.59]
cma	-26.19 [-4.09]	-7.73 [-1.06]	-9.61 [-1.33]	-26.61 [-3.91]	-6.71 [-0.92]	-7.35 [-1.03]	-31.13 [-4.13]
umd	-6.87 [-2.94]	-8.79 [-3.49]	-8.47 [-3.46]	-6.49 [-3.09]	-8.32 [-3.27]	-11.25 [-3.22]	-7.61 [-2.37]
# months	577	680	684	630	680	684	577
$\bar{R}^2(\%)$	27	22	22	29	22	22	28

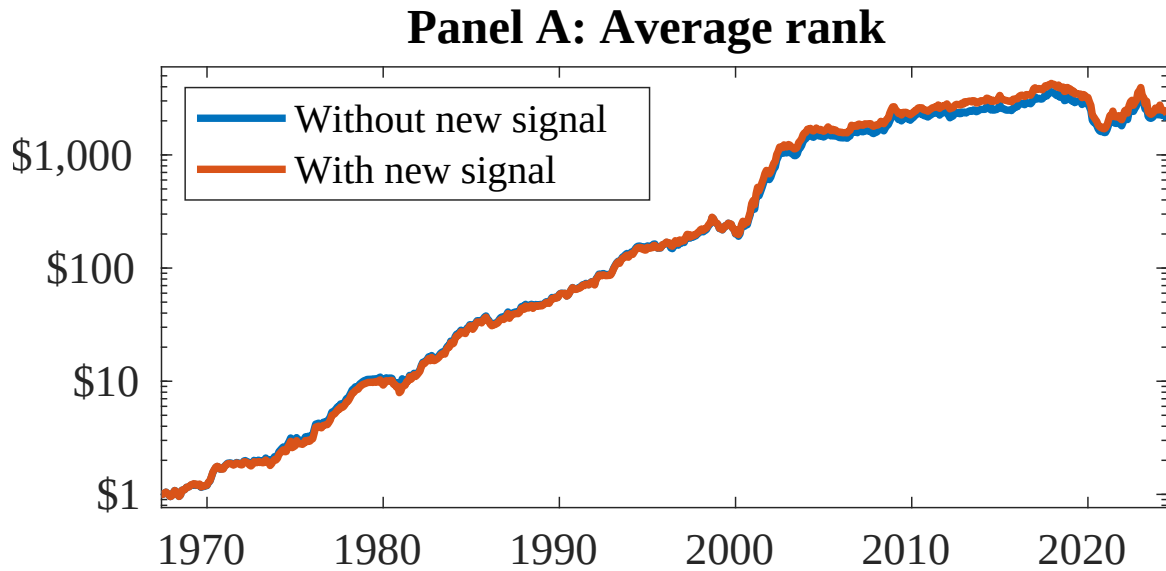


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 156 anomalies. The red solid lines indicate combination trading strategies that utilize the 156 anomalies as well as CCC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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